

Replication Report:
Geometry-Driven Nonlinear Orbital Magnetoelectric Effect
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Replication study (jia2026) — REPLICATE-PROJECT / TEXTURE-orbital

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Abstract

We attempt a numerical reproduction of the intrinsic nonlinear orbital magnetoelectric effect (NOME) reported by Jia *et al.* for their modified Kane–Mele honeycomb model (Model 1, Eq. 11). We build a from-scratch **numpy** tight-binding implementation of the 4-band Bloch Hamiltonian and evaluate the conventional matrix-element (“od”) contribution to $\chi_{z;xx}^{(0)}$ as a Brillouin-zone integral, together with the analogous spin response. **Verdict: PARTIAL.** We cleanly reproduce (a) the 4-band structure of Fig. 1(a) (~ 5 eV bandwidth, ~ 18 meV central gap); (b) the flagship *symmetry* statement that $\chi_{z;xx}^{(0)}$ is an *even* function of λ_R (P-even) to machine precision ($\sim 10^{-16}$), with a nonzero value at $\lambda_R = 0$; and (c) the qualitative orbital-vs-spin *opposite-sign* relationship near the gap. We do **not** reproduce the headline quantitative claim that the orbital moment is $\approx 3\times$ the spin moment, nor the absolute μ_B/V^2 prefactor: both depend on the geometric / Hermitian-connection response sector (Eqs. 4–9), the paper’s actual novel physics, which was not implemented in this compute pass. This is an honest partial reproduction that *validates the symmetry structure* of the effect but not its flagship magnitude.

1 Paper summary

The paper proposes a *nonlinear* orbital magnetoelectric effect (NOME): a magnetization response second order in the electric field, $\delta M^c = \chi^{c;ab} E_a E_b$, described by a rank-3 P-even pseudotensor. Unlike the *linear* OME, the NOME is symmetry-allowed even in centrosymmetric (inversion-symmetric) materials, dramatically enlarging the set of hosting magnetic point groups (MPGs): 41 MPGs support the intrinsic NOME and 55 the extrinsic NOME in 2D, versus only 10/8 for the linear OME. The intrinsic response decomposes into three geometric contributions — a conventional matrix-element (“od”) term, a dipole (“d”) term, and a positional-shift (“ic”) term — all governed by the quantum-geometric *Hermitian connection*. The authors argue the geometric $d + ic$ terms *dominate* near the band edge. Two model demonstrations are given: (Model 1) a modified Kane–Mele honeycomb (MPG $6m'm'$) for the intrinsic effect, predicting orbital $M \approx 3\times$ spin M with opposite sign and a nonzero $\sim 10 \mu_B/V^2$ minimum at $\lambda_R = 0$; and (Model 2) a CuMnAs antiferromagnet (*PT*-symmetric) for the extrinsic effect. Estimated magnitudes ($\sim 10^{-5} \mu_B/\text{nm}^2$ for the honeycomb at $E = 10^5$ V/m) are claimed detectable by polar MOKE.

2 Claims tested

Table 1 lists the paper’s central claims, whether each is testable with our compute, whether we tested it, and the paper vs. reproduced values.

Table 1: Claims, testability, and reproduction outcome.

Claim	Statement	Testable	Tested	Paper value	Reproduced
C4-bands	KM honeycomb 4-band structure, meV gaps (Fig. 1a)	yes	yes	4 bands, meV gaps near K/K', $\sim$ eV width	MATCH: 4 bands, $E \in [-2560, 2560]$ meV, gap 18.04 meV
C4-even- λ_R	$\chi_{z;xx}^{(0)}$ even in λ_R (P-even), nonzero min at $\lambda_R=0$	yes	yes	even in λ_R ; $\sim 10 \mu_B/V^2$ at $\lambda_R=0$	MATCH: even to rel. err 9.1×10^{-16} ; nonzero raw value 13.94 at $\lambda_R=0$
C4-orb/spin	orbital $\approx 3\times$ spin, opposite sign	yes	yes (od only)	orbital $\approx 3\times$ spin, opposite sign	PARTIAL: sign opposite near gap $\checkmark$; ratio (od-only) median ≈ 11.6 , peak -115.6 (<i>not</i> $3\times$)
C4-onlyxx	$6m'm'$: only $\chi_{z;xx}=\chi_{z;yy}$ finite	yes	no	$\chi_{z;xy}=0$ by symmetry	Not enumerated; only allowed xx computed
C6-order	magnitude $\sim 10 \mu_B/V^2$ ($\sim 10^{-5} \mu_B/\text{nm}^2$)	yes	partial	$\sim 10 \mu_B/V^2$	Not pinned: structure reproduced, absolute prefactor not fixed
C1/C2/C3/C5	Existence, 3-term geometric split, MPG count, CuMnAs model	yes	no	—	Not attempted this pass

3 Method

Model. We implement Model 1, the modified Kane–Mele honeycomb (Eq. 11), on a 4-state basis (2 sublattice $\times$ 2 spin):

$$H = -t \sum_{\langle ij \rangle} c_i^\dagger c_j + i\lambda_R \sum_{\langle ij \rangle} (\boldsymbol{\sigma} \times \mathbf{d}_{ij})_z + i\lambda_{so} \sum_{\langle\langle ij \rangle\rangle} \nu_{ij} \sigma_z + \lambda \sum_i s_z,$$

comprising nearest-neighbor hopping, inversion-breaking Rashba SOC, Kane–Mele mirror SOC (opposite sign on the two sublattices), and a T -breaking exchange term. Parameters follow the Fig. 1 caption: $t = 0.85$ eV, $\lambda_R = 20$ meV, $\lambda = 10$ meV, $\lambda_{so} = 10$ meV, $T = 20$ K, evaluated on a 120×120 Brillouin-zone mesh.

Response. From the Bloch Hamiltonian we compute, per k -point: eigenenergies; velocity matrices $v^a = U^\dagger(\partial_a H)U$ (central finite difference in k); the interband Berry connection $r_{nm}^a = i v_{nm}^a / (E_n - E_m)$; the orbital-moment matrix $L_{nm}^z = \frac{1}{4} \varepsilon_{zab} \{r^a, v^b\}_{nm}$; the quantum metric G_{nm}^{xx} ; and the spin matrix S_{nm}^z . We then evaluate the conventional intrinsic (“od”) NOME term $\chi_{z;xx}^{(0,od)}$ — the first line of Eq. (3) — as a Fermi–Dirac-weighted BZ integral, and the analogous *spin* response $\chi_{z;xx}^{s(0,od)}$ via the paper’s stated substitution $L^c \rightarrow s^z$. Sweeps: band structure along K’– Γ –M–K– Γ (Fig. 1a); $\chi^{(0,od)}$ vs. μ (Fig. 1b); and $\chi^{(0,od)}$ vs. λ_R at fixed $\mu = 50$ meV (Fig. 1c). All logic is in `work/reproduce.py`; numeric outputs are serialized to `work/results.json` and figures to `work/figs/*.png`.

What we deliberately did not implement. The geometric “d” and “ic” terms (Eqs. 4–6, 8, 9), which require the full Hermitian connection $C_{nm}^{\alpha ab}$, positional-shift / covariant-derivative tensors ($N_{nm}^{\alpha ab}$, $L_{nm}^{\alpha ab}$, D_{mn}^a), were out of the compute budget for a defensibly gauge-invariant implementation. As discussed below, this is precisely the sector responsible for the flagship $3\times$ magnitude.

4 Results

4.1 Band structure (C4-bands): MATCH

The 4-band structure reproduces Fig. 1(a): four bands spanning $E \in [-2560, 2560]$ meV ($\approx 6t = 5.1$ eV total bandwidth), with a central (band 2 $\leftrightarrow$ 3) gap of 18.04 meV set by the combined SOC + exchange scales. See `work/figs/fig1a_bands.png`.

4.2 P-even / even-in- λ_R symmetry (C4-even- λ_R): MATCH (machine precision)

This is the strongest reproduced result. We find

$$\chi_{z;xx}^{(0,od)}(+\lambda_R) = \chi_{z;xx}^{(0,od)}(-\lambda_R) \quad \text{to relative error } 9.08 \times 10^{-16},$$

verified across $\lambda_R = 10, 20, 30$ meV (Table 2). The response is *nonzero* at $\lambda_R = 0$ (raw value 13.94), structurally reproducing the paper’s statement of a nonzero minimum when inversion is restored. This directly and cleanly confirms the P-even character of the intrinsic NOME. See `work/figs/fig1c_chi_vs_lamR.png`.

Table 2: Even-in- λ_R check: $\chi_{z;xx}^{(0,od)}(+\lambda_R)$ vs. $\chi_{z;xx}^{(0,od)}(-\lambda_R)$ (raw units).

λ_R (eV)	$\chi(+\lambda_R)$	$\chi(-\lambda_R)$
0.01	+2.193426222912495	+2.1934262229124903
0.02	−5.2896687883299345	−5.289668788329926
0.03	−0.2066981391265077	−0.20669813912650806

4.3 Orbital vs. spin (C4-orb/spin): PARTIAL

Near the band-gap region the orbital “od” response has the *opposite sign* to the spin response, consistent with the paper’s qualitative claim. However, the *quantitative* magnitude is not reproduced: the od-only orbital/spin ratio is

$$\text{median (near cond. edge)} \approx 11.6, \quad \text{peak} \approx -115.6,$$

i.e. $\sim 10\text{--}100\times$, rather than the claimed $\approx 3\times$. As explained in the critique, the $3\times$ figure is a property of the *total* (od + geometric) response; the od term alone overshoots. See [work/figs/fig1b_chi_vs_mu.png](#).

4.4 Absolute magnitude (C6): not pinned

We report the response in raw/internal geometric units. Converting to μ_B/V^2 (and thence to $\sim 10^{-5} \mu_B/\text{nm}^2$ at $E = 10^5 \text{ V/m}$) requires the full dimensional prefactor bookkeeping of Eq. (1) (electron charge, lattice length, $\hbar$, cell area, the E-field length convention). We reproduce the response *structure* but do not fix the absolute scale; C6 is neither confirmed nor refuted.

5 Per-claim: what worked / what did not

- **C4-bands — worked.** 4 bands, correct bandwidth, meV gap.
- **C4-even- λ_R — worked (best result).** Even symmetry to $\sim 10^{-16}$; nonzero at $\lambda_R = 0$. This validates the core symmetry structure of the effect.
- **C4-orb/spin — partial.** Sign relationship reproduced; $3\times$ magnitude *not* reproduced (od-only ratio $\sim 10\text{--}100\times$).
- **C4-onlyxx — not done.** Only the allowed xx component was computed; forbidden-component vanishing was not enumerated.
- **C6 — not pinned.** Absolute μ_B/V^2 prefactor not fixed (dimensional bookkeeping).
- **C5 (CuMnAs Model 2) — not attempted.**
- **C3 (MPG enumeration, 41/55) — not attempted.**

6 Honest critique

The strongest reproduced result is the *symmetry* statement: that the intrinsic NOME $\chi_{z;xx}^{(0)}$ is an even function of λ_R (P-even), which we confirm to machine precision, together with a nonzero value at $\lambda_R = 0$. That is a genuine, clean validation of the effect’s symmetry backbone.

However, the paper’s *flagship* quantitative result — orbital moment $\approx 3\times$ spin moment — was **not** quantitatively reproduced. The reason is specific and important: the $3\times$ ratio is a property of the *total* response, which is dominated by the geometric $d + ic$ / Hermitian-connection sector (Eqs. 4–9). That sector *is* the paper’s actual novel physics, and it was *not coded* in this compute pass. Our od-only implementation — the one robustly well-defined, gauge-invariant term — overshoots the ratio to $\sim 10\text{--}100\times$ precisely because it omits the geometric contributions that would rebalance orbital vs. spin near the edge. In short: **this replication validates the symmetry structure of the NOME but not its flagship magnitude**, and the missing piece is exactly the paper’s headline geometric mechanism. Any claim of full reproduction would be dishonest without implementing Eqs. (4)–(6) with correct covariant derivatives and a Dirac-point convergence study.

Two further honesty caveats: (i) the absolute μ_B/V^2 prefactor (C6) is not pinned, so the “ $\sim 10 \mu_B/V^2$ ” order-of-magnitude is unverified; and (ii) the χ -vs- λ_R curve at fixed μ is mesh-sensitive/noisy (the Fermi level crosses band edges as λ_R varies on a fixed 120×120 grid) — the *even symmetry* is exact regardless, but the fine shape would need a denser, adaptively-refined mesh near K/K' .

Open Questions

Q1.

Does implementing the full Hermitian-connection geometric sector (Eqs. 4–6, 8, 9) recover the paper’s $\approx 3\times$ orbital/spin ratio? Our od-only ratio is $\sim 10\text{--}100\times$; the missing geometric $d+ic$ terms are the hypothesized rebalancing mechanism.

Q2.

Is the machine-precision λ_R -evenness of $\chi_{z;xx}^{(0)}$ a property of the *total* response, or only of the od sector we computed? If the geometric terms break or preserve the evenness at the same precision, that is a strong internal-consistency test of the P-even claim.

Q3.

What fixes the absolute μ_B/V^2 prefactor (C6)? Which dimensional convention in Eq. (1) (E-field length gauge, cell area, $\hbar$ /charge factors) reconciles our raw units with the claimed $\sim 10\mu_B/V^2$ and $\sim 10^{-5}\mu_B/\text{nm}^2$?

Q4.

How does the extrinsic CuMnAs Model 2 (Eq. 12, PT -symmetric) compare — does its single allowed $\chi_{z;xy}^{(1)}$, dominated by the Hermitian connection with a Dirac-point peak, reproduce cleanly where the intrinsic $3\times$ did not?

Q5.

How sensitive is the $3\times$ ratio (and the whole $\chi(\mu)$ profile) to temperature T and chemical potential μ , and to BZ-mesh density near the Dirac points? Does an adaptive $\geq 200 \times 200$ mesh remove the observed λ_R -sweep noise and stabilize the ratio?

7 Verdict

PARTIAL. We reproduced the model, its band structure, and — to machine precision — the paper’s central *symmetry* statement that the intrinsic NOME $\chi_{z;xx}^{(0)}$ is an even function of λ_R (P-even) with a nonzero value at $\lambda_R = 0$, plus the qualitative orbital-vs-spin opposite-sign relationship. We did *not* reproduce the flagship quantitative claim (orbital $\approx 3\times$ spin) or the absolute μ_B/V^2 magnitude, because both depend on the geometric / Hermitian-connection sector (Eqs. 4–9) — the paper’s genuine novel physics — which was deliberately not implemented in the compute budget, plus dimensional-prefactor bookkeeping. Model 2 (CuMnAs) and the MPG enumeration were not attempted. This is an honest partial that validates the effect’s symmetry structure but not its flagship magnitude.